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Ryan Campbell

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Research Interests

Multivariate and spatio-temporal extreme value theory, dependence modelling, applications to real world data.

Professional Activities

- **2025–2028** Postdoctoral researcher, Université Côte d'Azur, Nice, France
- **2020–2021** Data science intern, Desjardins General Insurance Group, Lévis, Québec

Education

- **2021–2025** PhD Statistics, Lancaster University
  Thesis: *A geometric interpretation of multivariate extreme value analysis*
  Supervisor: Jennifer L. Wadsworth
- **2019–2020** MSc Mathematics & Statistics, McGill University
  Thesis: *Deterministic Gaussian averaged neural networks*
  Supervisor: Adam Oberman
- **2015–2018** BSc Mathematics, McGill University
  Project: *Semiparametric modelling of max-stable processes using Kendall's tau*
  Supervisor: Johanna Nešlehová

Publications

Published/To Appear

1. **R. Campbell**, K. Grolmusova, L. Kakampakou, and J. Lee. Analysing Extreme Rainfall via a Geometric Framework, *Extremes. To appear. 2026.*
2. L.M. André, **R. Campbell**, E. D'Arcy, A. Farrell, D. Healy, L. Kakampakou, C. Murphy, C.J.R. Murphy-Barltrop, M. Speers. Extreme value methods for estimating rare events in Utopia. *Extremes*, 1–23, 2024.
3. J. L. Wadsworth and **R. Campbell**. Statistical inference for multivariate extremes via a geometric approach. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, (86)5: 1243–1265, 2024.

In Preparation

1. **R. Campbell**, E. Di Bernardino, and T. Opitz. Analysis of spatial extreme events via excursion set geometry. *In preparation*, 2026+.
2. **R. Campbell** and J. L. Wadsworth. The geometry of standardized multivariate generalized Pareto random vectors. *In preparation*, 2026+.

Preprints/Submissions

1. **R. Campbell** and J. L. Wadsworth. Piecewise-linear modeling of multivariate geometric extremes. *arXiv:2412.05195*, 2024.
2. I. Papastathopoulos, L. de Monte, **R. Campbell**, H. Rue. Statistical inference for radially-stable generalized Pareto distributions and return level-sets in geometric extremes. *arXiv:2310.06130*, 2023.
3. **R. Campbell**, C. Finlay, and A. M. Oberman. Adversarial Boot Camp: label free certified robustness in one epoch. *arXiv:2010.02508*, 2020.
4. **R. Campbell**, C. Finlay, and A. M. Oberman. Deterministic Gaussian averaged neural networks. *arXiv:2006.06061*, 2020.

Selected Presentations

- **June 2026** Extreme events in agriculture and forestry: Challenges, methods, applications
  Title: Analysis of extreme spatial dependence using excursion set geometry
  Location: Avignon University, Avignon, France
- **June 2026** 57$^{\text{es}}$ Journées de Statistique de la Société Française de Statistique
  Title: Analysis of extreme spatial dependence using excursion set geometry
  Location: Clermont Auvergne University, Clermont-Ferrand, France
- **June 2025** 14$^{\text{th}}$ International Conference on Extreme Value Analysis
  Title: Geometric spatio-temporal extremes *(with Kristina Grolmusova, Lydia Kakampakou, and Jeongjin Lee)*
  Location: University of North Carolina, Chapel Hill, NC, USA
- **June 2025** 14$^{\text{th}}$ International Conference on Extreme Value Analysis
  Title: Piecewise-linear modeling of multivariate geometric extremes
  Location: University of North Carolina, Chapel Hill, NC, USA
- **Nov. 2024** Various statistics seminars in France and Switzerland
  Title: A geometric approach to multivariate extremal inference
  Locations: Inria Grenoble, Université de Genève, École Polytechnique Fédérale de Lausanne, Université Claude Bernard Lyon 1, INRAE Avignon
- **Sept. 2024** UQÀM Statistics Seminar
  Title: New developments for a geometric approach to multivariate extremal inference
  Location: UQÀM, Montréal, QC, Canada
- **Oct. 2023** HEC Statistics Seminar
  Title: Statistical inference for multivariate extremes via a geometric approach
  Location: HEC Montréal, Montréal, QC, Canada
- **Sept. 2023** STOR-i Extremes Workshop (STEW)
  Title: Modelling extremal dependence of a 3-dimensional oceanographic dataset via a semi-parametric geometric approach
  Location: Lancaster University, Lancaster, UK
- **June 2023** 13$^{\text{th}}$ International Conference on Extreme Value Analysis
  Title: A geometric approach for modelling negative asymptotic dependence
  Location: Bocconi University, Milan, Italy

Teaching Assistant Positions

Lancaster University

- Winter 2024: MATH333 Statistical Models
- Winter 2024: MATH113 Convergence and Continuity
- Winter 2023: MATH140 Statistics

- Winter 2022: MATH456/556 Extreme Value Theory
- Winter 2022: MATH235 Statistics II
- Fall 2021: MATH330 Likelihood Inference

McGill University

- Fall 2020: MATH208 Intro. to Statistical Computing
- Fall 2019: MATH597 Topics in Applied Mathematics: Mathematics of Machine Learning
- Fall 2019: MATH223 Linear Algebra

Awards and Funding

- 2023–2025: NSERC Postgraduate Scholarship-Doctoral (CA$63,000)
- 2023–2025: FRQNT Doctoral Research Scholarship (CA$25,334)
- 2021–2025: EPSRC Mathematical Sciences studentship (£62,436 minimum)
- 2025: London Mathematical Society Early Career Researcher Travel Grant (£500)
- 2025: Institute of Mathematics Small Grant (£600)
- 2025: Graduate College & Doctoral Academy Travel Grant, Lancaster University (£200)
- 2023: Nick Smith Prize, Lancaster University (£500)
- 2020: Mitacs internship at Desjardins (CA$13,000)
- 2019–2020: Master's degree funding (CA$20,500)
- 2019–2020: McGill University Graduate Excellence Award (CA$3,400)
- 2018: Science Undergraduate Research Award (CA$6,500)

Languages and Skills

- Fluent in English and French
- Proficient in R, Python (incl. PyTorch), Matlab, LaTeX, Java, HTML, Linux

Extracurricular Activities

- **Treasurer**, Bowland Ultra, Lancaster, UK (2023–2025)
- **Treasurer**, Lancaster University Folk Society (2023–2025 academic years)
- **VP Finance**, Graduate Student Association for Mathematics and Statistics (GSAMS), McGill University (2019–2020 academic year)
- **Volunteer** at the 2018 Statistical Society of Canada annual meeting (3–6 June 2018, McGill University)
  Roles: Setting up audio-visual equipment and directing conference attendees